2.3 Epithelial Cells and Tissues

- What are tissues?
  - A group of cells whose type, organisation and architechture are integral to its function.
- What is the epithelium?
  - The type of tissue that forms the covering on most internal and external surfaces.
- What are epithelial cells?
  - The major cell type in living organisms, that are involved in many biochemical and physiological processes in humans.
  - They **line the surfaces of structures**
- What are the three things tissues are made of?
  - **Cells**
  - **Extracellular matrix**
  - **Fluid**
- What is the extracellular matrix?
  - Material deposited by cells
  - It forms the **insoluble part of the extracellular environment.**
- What is the extracellular matrix made of?
  - Fibrillar/reticular proteins embedded in a hydrated gel
- Recall the five main cell types and the areas of the body they are associated with.
  - **Connective tissue cells** - bone
  - **Contractile tissues** - muscle
  - **Haematopoietic cells** - blood
  - **Neural cells** - nervous system
  - **Epithelial cells** - cells that line surfaces
- What are epithelial cell cancers called?
  - **Carcinomas**
- What are connective and contractive cell cancers called?
  - **Sarcomas**

▼ What are neural cell cancers called?

- **Neuroblastomas**
- **Gliomas** for glial cells

▼ What are haematopoietic cell cancers called?

- **Leukaemias** - all blood cells in circulation
- **Lymphomas** - lymphocytes in blood circulation

▼ How are epithelial cells adapted to perform their function?

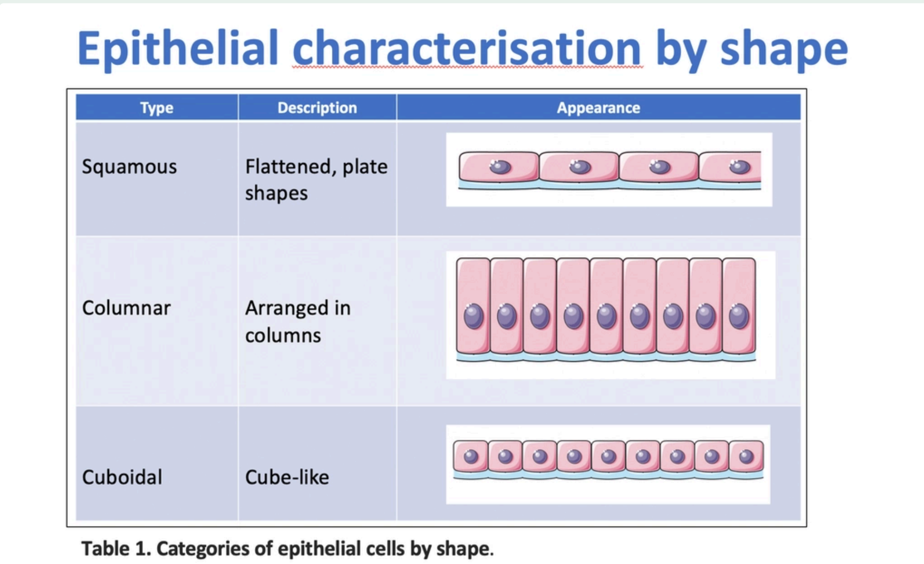
- They form stable cell-cell junctions that help them to perform their functions efficiently.

▼ Describe the different functions that epithelial layers have.

- Transport
- Absorption
- Secretion
- Protection

▼ Recall the three shape classifications of epithelial cells.

- **Squamous** - flattened
- **Cuboidal** - cube-like structures
- **Columnar** - tall + long



Epithelial characterisation by shape

Type	Description	Appearance
Squamous	Flattened, plate shapes	
Columnar	Arranged in columns	
Cuboidal	Cube-like	

Table 1. Categories of epithelial cells by shape.

- Recall the two layering classifications of epithelial cells.
  - **Single-layer (simple)**
  - **Multi-layer (stratified)**

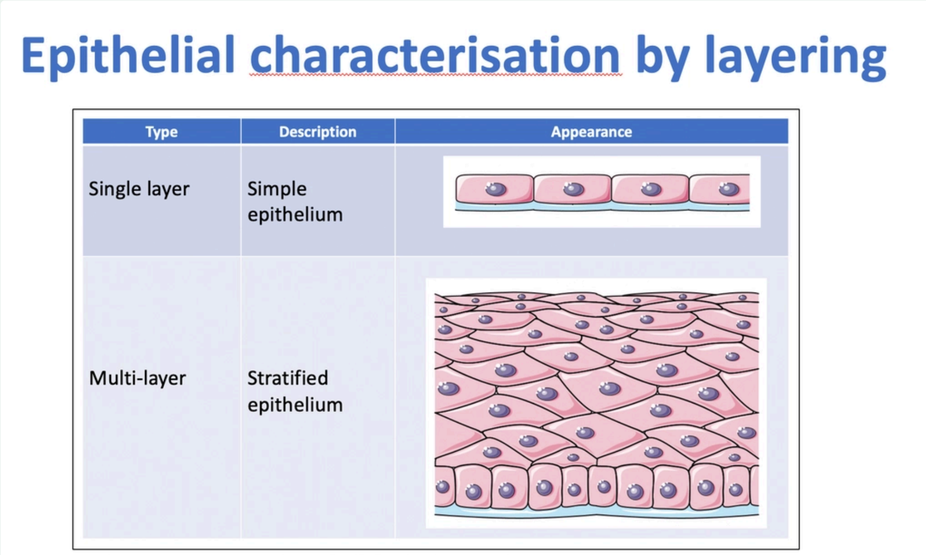


Table 2. Categories of epithelial cells by layering.

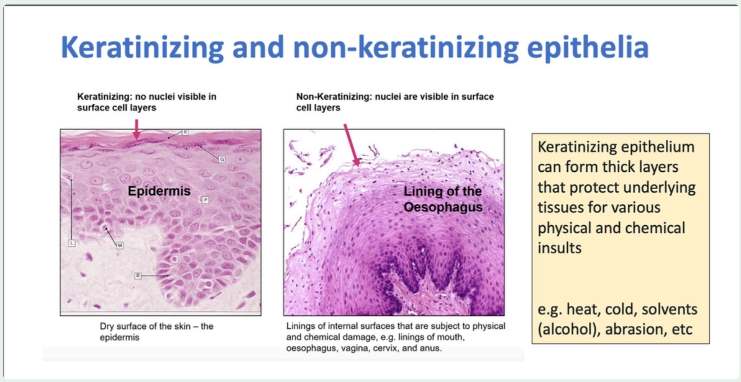
- Where are simple squamous epithelial arrangements found and why?
  - In the **lung alveolar epithelium**
  - They are the **thinnest types of epithelial cells** so this makes gas exchange efficient
- Where are simple cuboidal epithelial arrangements found?
  - **Ducts** (circular channels leading from glands or organs)
- Where are simple columnar epithelial arrangements found?
  - In surfaces where there is a need for **absorption and secretion of molecules (i.e intestine)**
- What are the two classifications of stratified squamous epithelium?
  - **Keratinizing**
  - **Non-keratinizing**
- What is the difference between these two types of epithelium?
  - Keratinizing epithelium **produce keratin** which causes them to **die** and instead **become thicker**, **protective structures**
  - Non-keratinizing epithelium **do not produce keratin**, so they **remain in tact with all their organelles.**
- Give an example of a keratinising stratified squamous epithelium layer.
  - The **epidermis (skin)**

▼ Describe an adaptation that allows keratinising epithelium to perform their functions.

- They can form **thick layers** that help **protect the underlying tissues** from various types of damage (i.e heat, cold)

▼ What is the difference between keratinising and non-keratinising epithelia when viewing them under microscopes?

- **Keratinising epithelia** lose their nuclei and organelles when they produce keratin, so they do not have these structures visible under a light microscope
- **Non-keratinising epithelia** retain their nuclei and organelles, so they do have these structures visible under a light microscope



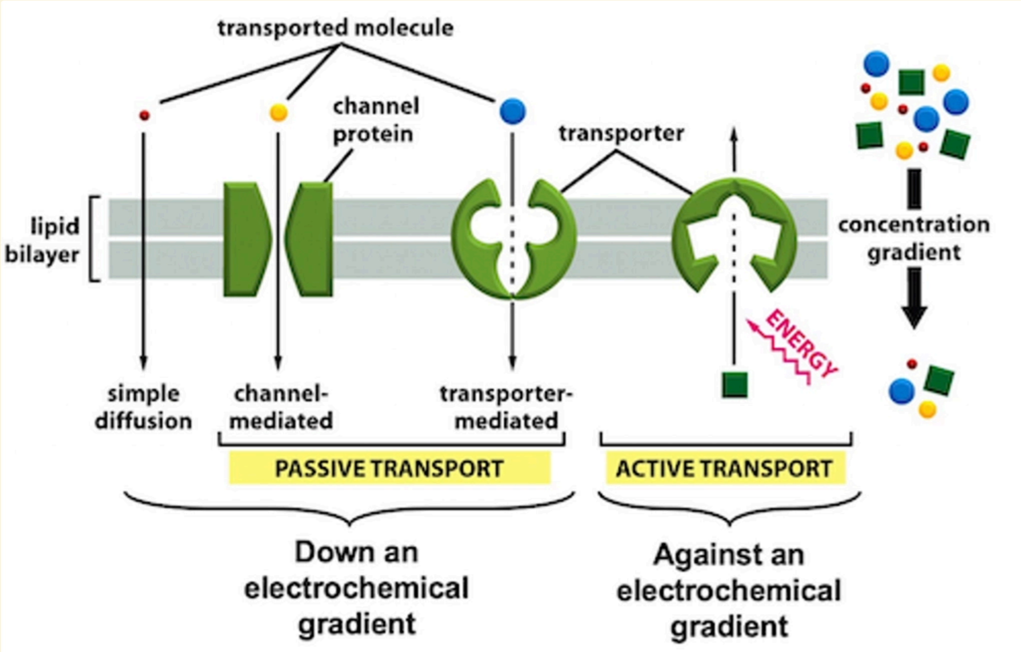
▼ Describe what pseudo-stratified epithelia are.

- Those that appear to be multi-layered, but upon closer examination **can be seen to touch the basal lamina**
- **“Fake-stratified”**

▼ What usually drives the “directionality” seen in epithelial cells that allows them to perform their functions (i.e transporting, secreting)?

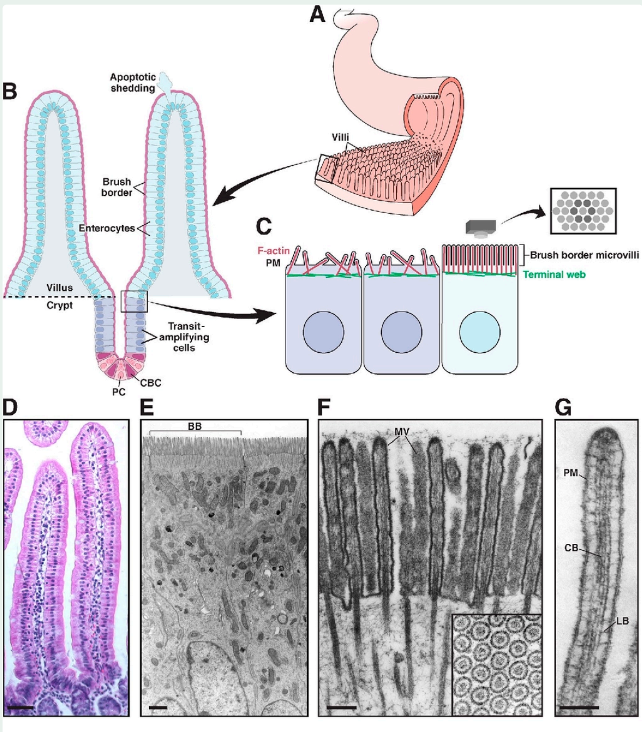
- **Plasma membrane polarity.**

- What is present between the lipid bilayers of epithelial layers?
  - **Proteins**.
    - i.e **transporter proteins, channel proteins**



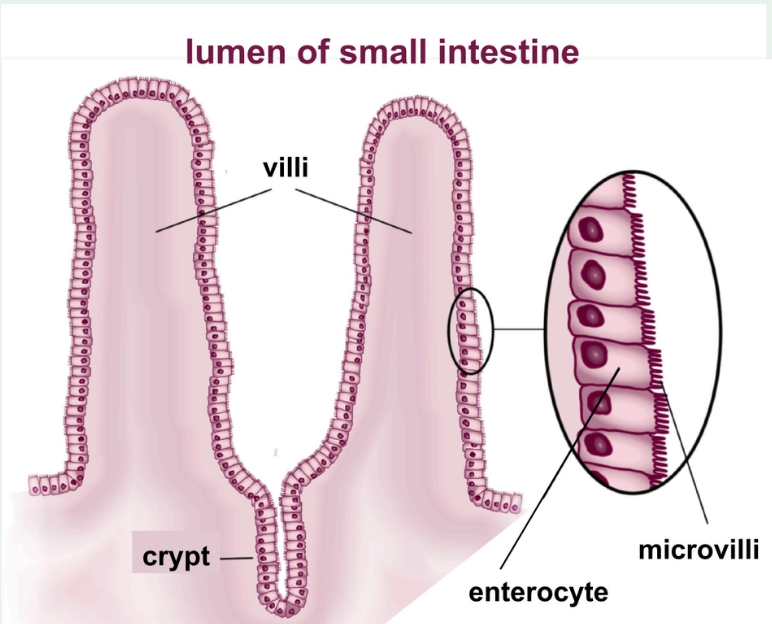
- What is Na+/K+ ATPase and what is it used for?
  - It is an enzyme which acts as a protein pump for the movement of Na+ and K+ across cellular membranes
  - It moves these ions against concentration gradients, hence it requires ATP for its function.
  - It is used to **maintain the plasma membrane potential**
- How does plasma membrane polarity help epithelial cells?
  - It can help with **directionality**
  - Those that are involved in secretion or transport for example, can **use polarity to cause a desired net movement** of particular molecules in a particular direction
- When is ATP hydrolysis required for transmembranal transport?
  - When substances are being transported **against a concentration gradient.**

- Describe two structural adaptations that transporting epithelium have which allows them to perform their function.
    - They have **high concentrations of ion transporters in their membranes** which allow them to transport substances in large volumes
    - They have **mitochondria closely associated with them** which provide energy for pumping
    - They have **membrane infolds** that allow them to pump as many substances as possible.
- Where are the mitochondria present in transporter epithelial cells located?
    - They are located towards the **basal side of the cell,** to aid with the **movement of ions and other molecules into and out of the blood**
- What is the brush-border membrane?
    - The **entire area on an epithelial cell characterised by its plasma membrane that has folded into its microvilli structure**



- What are enterocytes?
    - **Cells** that are found in the **intestinal lining**
- What are goblet cells?
    - **Intestinal epithelial cells** that **produce mucus**.

- Describe a distinctive feature of the brush-border which allows them to perform their functions.
  - They are **rich in active transporters and channels** which allow the **uptake of nutrients** from the gut lumen
- What is the difference between villi and microvilli?
  - **Villi** are the finger-like projections that are formed by groups of epithelial cells that line tissues (i.e small intestine) to increase the surface areas of the tissues involved
  - **Microvilli** are the finger-like projections that are formed on the plasma membranes of individual epithelial cells that line these cells to increase their surface areas
  - Both are adaptations used to increase surface areas for diffusion, but microvilli are found on individual cells, villi are found on tissues



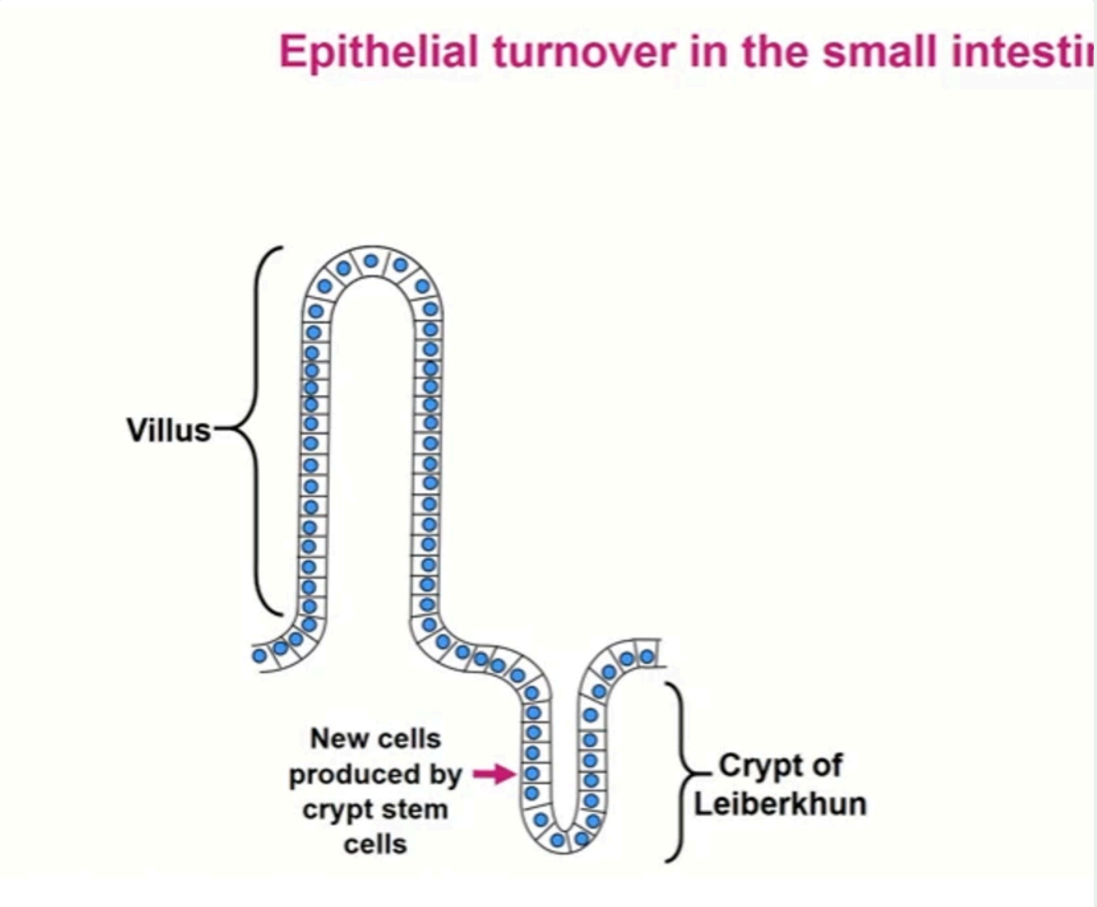
- Describe the difference between exocrine and endocrine secretion.
  - **Exocrine** secretion involves the release of substances **into a duct or lumen**
  - **Endocrine** secretion involves the release of substances **into the blood.**
- What are the two ends of an epithelial cell known as?
  - **Apical**
  - **Basal**
- Which end does secretion occur from for exocrine secretory cells?
  - **Apical**.

▼ Which end does secretion occur from for endocrine secretory cells?

- **Basal**.
- This is because the **basal membrane is located closer to the capillaries** and so the vesicles secreted by the epithelial cells can access the bloodstream more easily.

▼ What are intestinal crypt cells?

- Cells which are present in the **intestinal crypt**
- The crypt is vital in the **proliferation and replacement of intestinal epithelial cells** after they die



▼ How do epithelial cells in the intestine proliferate after damage?

- **Cells from the intestinal crypts are produced** and **rise to the surface** where the damage/cell death has occurred